

MISTYCAT

SPACE X

SPATIAL INTELLIGENCE

STEREO IMAGING, TUNED BY EAR

Manual



Version 1.0 · VST3 · macOS

What this is for

SpaceX is not a corrective tool. It is a way to give a sound its own layer in a mix that is already full.

Most stereo plugins are built to be safe. This one is built to be interesting. It moves things sideways, forwards, backwards and slightly out of phase - the area most engineers stay out of, either because nobody ever showed them how far it can be taken, or because they are afraid of what it does to mono. Used deliberately, that is exactly where the space is.

That comes out of the work. I have been mixing professionally since 2019 as a producer, mix and mastering engineer, on records and for television, among them The Voice. Alongside that I have been hired to rebuild and optimise recording and mixing templates for other engineers - Jeff Ellis and Mosty among them - so that their way of working would run on a different system. Sitting inside someone else's session is the fastest way to see where the decisions really are.

Two terms from Jeff Ellis' Mixer Brain course stuck, because they put a name on something I had been doing for years without one: the forbidden zone, and the fourth dimension. When a mix is dense and every frequency is taken, width and depth are the only room left. SpaceX is built to work in that room.

Think of it as an effect generator rather than a mastering tool. It takes something ordinary - a reverb return, a delay throw, a stack of backing vocals - and makes it worth listening to.

What it replaces

Everything below normally needs four or five separate plugins:

Centre extraction (L / C / R)	a dedicated centre-separation plugin
Haas delay and micro pitch	a pitch / delay tool
Polarity, side gain, side width	a utility plugin - usually an ugly one
Mono check and correlation	a metering plugin
Depth and distance	an EQ move you would otherwise do by hand
Auto-pan and stereo movement	a panner
Stereo phaser	a modulation plugin

One window, one latency, one preset. That is most of the point.

Where to put it

It works best on anything that is already wet: backing vocals, ad-libs, a stack of harmonies, a reverb or a delay return. Throw tracks belong here too - those are usually single tracks rather than buses. A dry source is not off limits either: a lead vocal that is meant to be doubled works, and so do drums. The rule of thumb is simple - the more room effects sit in front of SpaceX, the further you can push it.

Position in the chain is a taste decision. Last is the usual place: shape whatever comes out of the chain. First also works when the incoming signal is already drenched in reverb or echo - then SpaceX shapes the effect itself rather than the source.

Throw tracks are the easiest win. They are 100 % wet by definition, nobody expects them to be mono-safe, and a little Orbit with some Parallax turns a plain delay into a hook.

It is not a mastering tool. Do not put it on the mix bus and expect it to behave.

First result in one minute

1. Put it on a backing vocal bus or a throw track.
2. Click the profile pill at the top until it says what the track is - Vocal, Backing, Adlib or FX.
3. Click the dice next to it. Listen. Click again. Keep going until something surprises you.
4. Too much? Pull Mix down to 70 %. Still too much? Click a section name to switch it off.
5. Check Mono at the bottom. If it collapses, lower Size in Mid-Side a little.

That is the whole workflow. The knobs below exist for when you already know what you want.

The Smart engine

The dice at the top builds a complete setting for you - and decides which sections belong in it. A section that would get nothing useful is switched off rather than left sitting there doing nothing, so you can always see what the roll actually did.

The pill next to it says what you are working on. Five positions: No Profile, Vocal, Backing, Adlib, FX. Click for the next one, Cmd-click to go back, or click a dot to jump straight to one. With a profile selected the dice stays inside what makes sense for that source; the line above the field says what the profile does.

Press it, listen, press it again. Keep going until something surprises you - and it will.

Don't worry about what it might do to your track. This is not the randomise button other plugins have. Everything it can reach was tuned by ear, and whatever did not work never made it in - so you will not get a result you have to undo.

Profile pill	Vocal, Backing, Adlib, FX - or No Profile, where the dice may reach for anything
Dice	rolls a new setting and decides which sections belong in it
Lock (padlock)	this section is left alone by the dice and by Breathe
Breathe	switches modulation on in every section with fresh depths
Life	scales every modulation at once. At zero nothing moves
Mod	switches all modulation off without losing the depths
Undo / Redo	works on everything, the dice included
A / B	two versions side by side; the arrow copies one into the other

The signal path

Left to right, top to bottom - the way you read it is the way the audio travels.

IN

GALAXY

ECLIPSE

PARALLAX

DIMENSION

HYPERDRIVE

RAYE

MIX

OUT

Eclipse is the exception: it can sit at two points in that chain. Early flips right after Galaxy, Late at the very end. The button shows where it currently sits.

Latency is zero unless Galaxy is armed. Galaxy needs an analysis window to separate the centre, so switching it on costs a little delay. Everything else runs in real time - which is why the engine has its own switch in the header rather than being always on.

On every section

Section name	click it to switch the section on or off
Cmd + name	solo - hear this section on its own
Padlock	the dice and Breathe leave this section alone
Wave icon	modulation on / off for this section
Small knob	modulation depth, scaled by Life

GALAXY

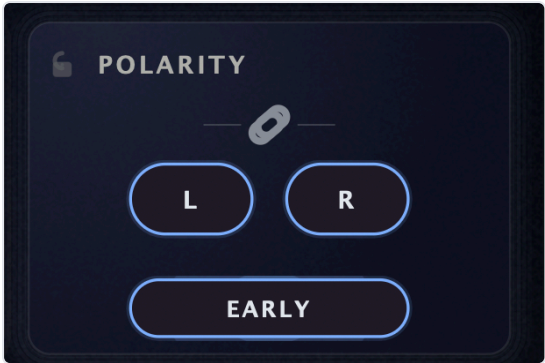


Pulls the centre out of the stereo image and treats centre and sides separately. This is the part that costs latency, and the part that makes everything after it sound different. It only runs while the engine is armed - the GALAXY button in the header.

Orbit	how much of the sides comes back in. All the way down is centre only
Gravity	how hard the centre is separated from the sides
Regain	brings back the level the separation takes away

On a backing stack, pull Orbit up. The lead stays where it is and the backings open up around it without getting louder.

ECLIPSE



Flips the polarity of one channel. The most drastic thing in the plugin and the one that makes the biggest difference.

L / R	flip the left or the right channel
Link	both at once
Early / Late	where in the chain the flip happens - click to switch

A flip almost always breaks mono compatibility. On a throw or an effect return that is fine - nobody sums those to mono and expects them to survive. On anything that carries the song, check Mono first.

The two positions matter more than they look. Flipping before the width stage changes what everything after it is working on; flipping at the end only affects the final image. Try both on the same setting - they are two different sounds.

PARALLAX



Turns a mono-ish sound into a wide one, using tiny differences between left and right in time and in pitch. One dial and one decision: pick a style, then set Amount.

The styles are finished settings, not presets you have to dial in. Amount runs from nothing to the full effect, and the dots under the button jump straight to a style.

Double	the plain doubler. Widest at full Amount, still solid in mono
Wide	a touch of detune with it - more air, less obvious doubling
Illusion	the big one. Delay and detune together, clearly out front
3D	leans the image and pushes it outward. Good under a lead
Drift	the same idea tilted to one side - asymmetric on purpose
Flux	a macro: starts small and ends at the full effect

Under the hood: Drift is a Haas delay of a few milliseconds, Shift a micro pitch offset in cents, Tilt a pan of the widened signal. Technical Labels renames the section MicroPitch.

DIMENSION



Classic mid / side width, plus depth. The section you reach for when the result is nearly right and just needs to be wider, heavier or further away.

Size	stereo width. Below 100 % narrower, above 100 % wider
Boost	adds level to the sides without touching the centre
Depth	left moves the sound back into the room and darkens it, right pulls it close

Size above roughly 150 % together with a strong Parallax style is where mono starts to suffer. The plugin already limits the worst combinations, but your ears decide.

HYPERDRIVE



Slow automatic movement through the stereo field. Not a tremolo - the level stays put, only the position moves.

Flow	how far the sound travels left and right
Sine / Pulse	the shape of the movement. Click to switch
Speed	movement rate in Hz
Sync	lock the rate to the host tempo
Bars	the note length used while Sync is on

RAYE



A phaser that moves the stereo image rather than the tone. Subtle on purpose - it is movement, not an effect you should be able to name. Amount is the whole control; the character decides how it moves.

Amount	how strong the movement is. At zero the section is silent
Sweep	the classic slow sweep, low and wide
Shimmer	high and quick - air rather than body
Spin	low, fast and thick. The obvious one
Swirl	very slow and broad. You notice it after a while, not straight away
Fast	runs the current character about a third quicker
Pair	follow Hyperdrive at half its speed, never faster than half a bar

Output



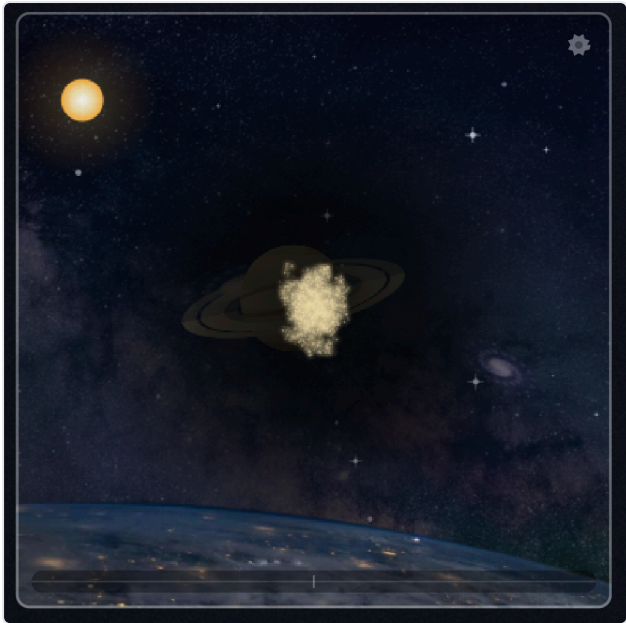
In / Out	input and output level
Mono	listen to the result in mono
Dry	while in mono, compare with the unprocessed input
Mix	blend the processed sound with the original
Vol	output trim, plus or minus 6 dB
Correlation	the thin bar in the field: right of centre is mono-safe, left of it cancels
AG	auto gain - matches the output to the input so bypass is an honest comparison

Modulation and Life

Every section can breathe. The wave icon in a section header switches its modulation on, the small knob next to it sets how far it moves. Switch Show Advanced Modulation on in the settings and both appear in every section; leave it off and the sections stay clean.

Life at the top scales all of it at once. At zero nothing moves at all, no matter what the individual depths say - which makes it the fastest way to calm a setting down without losing it. The moving dots on the knobs show what the modulation is doing right now.

The starfield



The picture on the left is not decoration only. Every object reacts to what the plugin is actually doing - the planet rises with Gravity, the blue lines come from Regain, the field curves with Orbit, and the whole scene leans with the movement.

Click the field to switch the scope on or off. The gear opens the display settings.

Goniometer	the classic vectorscope - a vertical line is mono, a circle is wide
Scene	Space shows everything, Stars is the quiet meter view
Speed / Density	how fast and how busy the field is
Stars / Lines	switch the twinkling stars or the warp lines off

Presets

The name field in the header shows what is loaded; the arrows step through the list. A star after the name means you have changed something since loading it.

Save	store the current settings. Name it Default to overwrite the default
Delete	remove the loaded preset
Rename / Preset Folder	in the preset list, below the names
Save State as Default	in the settings: every new instance starts like this

Presets that use Galaxy are greyed out while the engine is off, and the arrows skip them. Nothing loads a setting that cannot be heard.

Themes, layout and labels

Five themes and two layouts. Layout changes how much framing the sections get: 3D has soft shading and depth, Outline a thin frame and no fill. None of it changes the sound. Hover a theme to preview it, hover a layout to see what the framing does.

Technical Labels renames everything by what it does rather than what it feels like. Useful if you would rather see the engineering than the story:

Galaxy	LCR · Orbit becomes L/R, Gravity becomes C-Weight
Eclipse	Polarity
Parallax	MicroPitch
Dimension	Mid-Side · Size becomes Width, Boost becomes Gain
Hyperdrive	Autopan · Flow becomes Width
Raye	Phaser

Finding your way

? bottom left	switches the hint line on. It says what the mouse is over
Take the Tour	a short guided walk through the sections. In the settings
Back Panel	click the logo: who built this, how to reach him, which copy this is
i next to the profile	hides the Smart profile line if you do not want it

Demo and activation

Without a serial number SpaceX runs in demo mode: everything works, but the sound goes quiet for a moment every 50 seconds. The DEMO badge next to the logo lights up while that happens, so it is never a mystery. Click it and you can enter a serial, open the shop, or carry on.

Activation asks for your name and your serial number. The two belong together - the number is derived from the name, and the plugin checks them against each other. Type the name exactly as it appears in your order; upper and lower case and spaces do not matter. Your name then sits on the back panel under Registered to.

Five settings that work

Backing stack that will not fit

Backing profile, then the dice. Orbit up, Size around 120 %. The lead keeps the centre, the backings take everything around it.

Delay throw with an identity

Parallax on Illusion, Amount around 60 %, Raye on Sweep at a third. The throw stops sounding like a delay and starts sounding like a decision.

Reverb return with depth

Depth to the left, Size around 130 %, Boost a touch. The reverb moves behind the dry signal instead of sitting on top of it.

Ad-libs that appear and disappear

Hyperdrive Flow around 40 %, Sync on, 4 bars. Slow enough that nobody notices the movement, fast enough that the part never sits still.

The forbidden one

Eclipse L on, Early, Mix around 60 %. Mono will not survive it. On a throw in a dense chorus it is the widest thing you own.

Clicks worth knowing

Double-click a knob	back to default
Cmd / Ctrl + drag	fine adjustment
Click a section name	section on / off
Cmd + section name	solo that section
Click a style pill	next style. Cmd-click goes back, a dot jumps straight there
Click the logo	back panel
Right-click a knob	lock it against the dice

Credits

SpaceX started as a way to stop opening five plugins for one idea. Nothing in it happened by accident: the order of the chain, where the polarity flip is allowed to sit, how far each section may go - every one of those is a decision that was made by listening, and then made again.

Thanks to Jeff Ellis, whose words for the forbidden zone and the fourth dimension gave the idea a name. And to everyone who tested SpaceX and sent feedback - most of what is good about the last hundred details came from them.

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